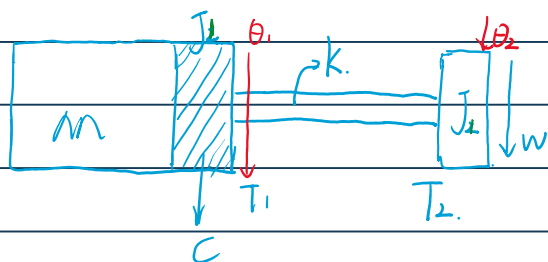


② 模型:

$f(t) = kx(t)$ 弹簧	电阻 $u(t) = i(t)R$
$f(t) = D\dot{x}(t)$ 阻尼	电容 $u(t) = \frac{1}{C} \int i(t) dt$
$f(t) = M\ddot{x}(t)$ 质量	电感 $u(t) = L \frac{di(t)}{dt}$
$T(t) = k\theta(t)$	
$T(t) = C\dot{\theta}(t)$	
$T(t) = J\ddot{\theta}(t)$	

m : 质量
 J : 转动惯量



$T_2 = k(\theta_1 - \theta_2)$ 弹性刚度体.

$J_1\ddot{\theta}_1 = T_1 - C\dot{\theta}_1$ $(J_1 + J_2)\ddot{\theta} = T_1 - C\dot{\theta}$

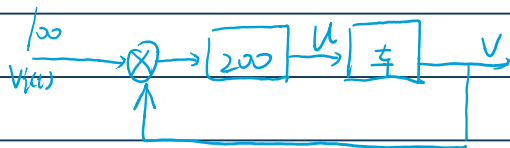
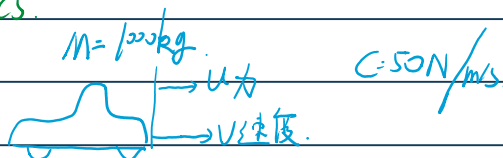
$J_1 s^2 \theta_1 = \frac{T_1}{s} - C s \theta_1$

$\textcircled{M} = T_1 / (sJ_1 s^2 + Cs) = \frac{T_1}{J_1 s^3 + Cs^2}$

$J_2 s^2 \textcircled{M} = K(M_1 - M_2) = K T_1 / (J_1 s^3 + Cs^2) - K M_2$

$M = \frac{T_1}{J_1 s^3 + Cs^2}$

③ 不要忘记输入



$Ma = U - Cv$ $M\dot{v}(t) = U(t) - C v(t)$

$M s V = U - \frac{C}{s} V$

$(sM + \frac{C}{s}) V = U$

$V = U / (sM + \frac{C}{s}) = \frac{U}{sM + C}$ $\therefore C(s) = \frac{200U}{sM + C} + 1$

$C(s) = \frac{200U}{200U + (sM + C)}$

二 系统时域响应

① 一阶: $\frac{k}{Ts+1}$ or $\frac{k}{s+a}$ 惯性环节.

快速. 估计. 近似. 画特征. 1) 稳态响应

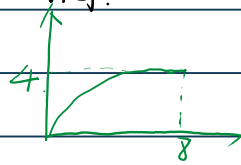
快速. 估计. 近似. 画特征. 1) 稳态响应.

2) 响应时间.

例 $\frac{8}{4s+2}$. $CSS = \frac{8}{2} = 4$.

$t_s = 4T \Rightarrow T=2$ $t_s = 8$.

$\frac{K}{s+a}$ $CSS(x) = \frac{K}{a}$. $t_s = \frac{4}{a}$.



5% $t_s = 3T$

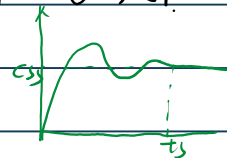
2% $t_s = 4T$.

② = 阶: $\frac{K}{T^2 s^2 + 2\zeta T s + 1}$ or $\frac{K}{s^2 + 2\zeta \omega_n s + \omega_n^2}$ $0 < \zeta < 1$.

t_s 48% $\approx \frac{4}{\omega_n \zeta}$

稳态 $CSS = \frac{K}{\omega_n^2}$.

超调量 $p_o = e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$



上升时间 $t_r = \frac{\pi - \arccos \zeta}{\omega_n \sqrt{1-\zeta^2}}$

峰值时间 $t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$

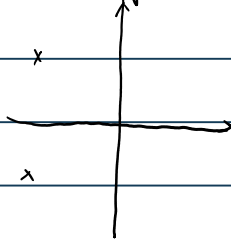
调整时间 $t_s = \frac{4}{\zeta \omega_n}$ $\frac{3}{\zeta \omega_n}$
 $\zeta=2$ $\zeta=5$

③ 有在主导极点 - 一般系统.

t_s 基于主导极点. 稳态 $CSS = \frac{K}{\omega_n^2}$

例 $\frac{8}{(s+2)(s+1)}$. $t_s = \frac{4}{2} = 2s$. $CSS = \frac{8}{2 \times 1} = 2$ 近似系统. $\frac{2 \times 1}{s+2}$ CSS 类似!

④ 极点与时间响应



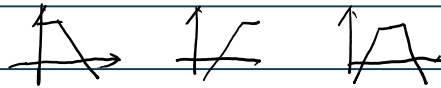
极点在复平面虚轴左侧有虚部 $\Rightarrow$ 稳定.

极点在复平面虚轴右侧有虚部 $\Rightarrow$ 不稳定.

奇振荡.

三 系统的频率特性.

① 滤波器 - 伯特图



② 伯德图 - 拆.

画 找起点

有稳态增益.

无稳态增益.

$\frac{(s+2)(s+5+10j)}{s^4 + 5s^3 + s^2 + 5}$ 分子/分母常.

$\frac{s(s+5)}{s^3 + 12s^2 + 4s + 7}$

$s = j\omega$. $\omega = 0.01$ 代入 $\Rightarrow$ by Krigum!

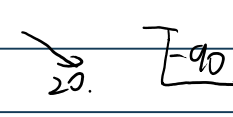
找转折频率.

$\frac{1}{s+a}$

$\omega T = a$

$\frac{1}{Ts+1}$

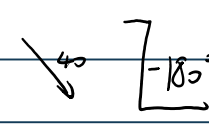
$\omega T = \frac{1}{T}$



$\frac{s^2 + 2\zeta \omega_n s + \omega_n^2}{T^2 s^2 + 2\zeta T s + 1}$

$\omega T = \omega_n$

$\omega T = \frac{1}{T}$



四 稳定性

① 分母不能缺项. 符号相同.

K^* 太大. 不稳定.

② PID 控制器

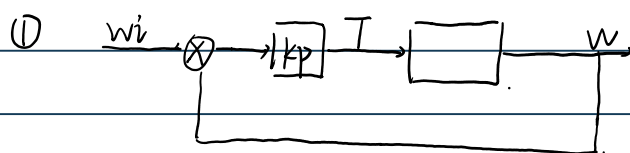
优 缺

k_p 基础 不稳定/振荡

k_i 消除稳态误差 振荡

k_d 减小振荡 高频噪声敏感.

五大题.



$$J = \frac{1}{3} m r^2$$

⑤ 4x5 选

4x5 填

^{2x15}
10 作图 → 阶跃响应

20 伯德图

30. 综合题 → 系统建模 反馈回路